

# ALVIN C. ADEFUIN

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AI developer who takes generative AI systems from prototype to production, and keeps them running afterwards. The retrieval and LLM work, the service around it, the front end, and the AWS infrastructure underneath. Now at Cambridge University Press & Assessment's Generative AI Centre of Excellence. Before that, shipped a RAG chat product that handled 4,500+ conversations at 29% lead conversion, and semantic search across 40,000+ SKUs that cut response times 3-4x.

## PROFESSIONAL EXPERIENCE

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### AI Software Developer

Feb 2026 - Present

Cambridge University Press & Assessment | Makati City

Working within a Generative AI Centre of Excellence, taking proof of concepts from prototype through to production services on AWS.

#### Marketing Persona Chatbot

- Built the FastAPI platform that answers questions from a vector store, scores its confidence in every answer, and routes the uncertain ones to a human reviewer. Small talk skips retrieval, so it doesn't burn an LLM call on "hi".
- Built the Next.js front end: sign-in with protected routes, admin screens for users and settings, and a chat interface that shows its sources and flags the answers it isn't sure about.
- Added and hardened sign-in: Microsoft Entra ID, JWT sessions, role-based access, and the CSRF, CORS and cookie handling that goes with it.
- Provisioned and ran the AWS infrastructure in Terraform: ECS Fargate, Aurora PostgreSQL, API Gateway and secrets. GitHub Actions pipelines gate changes behind review, run migrations and scan dependencies.

#### Image Deduplication Pipeline

- Contributed the image-similarity linking to a pipeline that groups duplicate submissions so the same thing isn't handled twice: working out which stored records refer to the same image, and the thresholds that decide when two are close enough to link.
- Added failure containment so a crashed batch, or one bad image group, fails on its own instead of stranding the rest of the run.

#### Document Processing Service

- Built a Go service that moves uploaded documents through a priority queue, with retry and dead-letter handling so nothing is lost or processed twice. Gin and pgx over PostgreSQL, with S3 and SQS, across 12 versioned migrations.

#### Cloud Governance Platform

- Contributed React features to an internal AWS governance tool: lease provisioning and experiment-attachment flows, recovery for deployments left stuck mid-provision, plus regression tests and accessibility fixes.
- Documented its API in OpenAPI and added a CI check that blocks pull requests when the docs drift from the code.

## AI Software Developer

Journey Better Business Group Inc. | Makati City

Aug 2023 - Jan 2026

### AI Chat Widget

- Built and shipped a RAG chat widget end to end, with a React/Preact front end and a Flask backend. 4,500+ conversations from 3,000+ unique users, 1,800+ qualified leads, 29% conversion.
- Took it multi-tenant across US and EU for 9+ client organizations, with Retool dashboards for analytics and Sentry for error tracking.
- Ran 47,000+ LLM calls against a knowledge base of 1,200+ chunks across 230+ documents, with citations and file upload.

### Natural Language Product Search

- Built semantic search across 40,000+ SKUs for a medical equipment distributor, cutting response times 3-4x.
- Built the FastAPI and Typesense backend that classifies complex scientific queries by category and pulls the filters out automatically.
- Built the TypeScript/Next.js interface, including a debug view showing exactly how a query was interpreted, model-number normalization, and category-specific brand ranking.

## Web Developer Intern

Highly Succeed Inc. | Remote

Feb 2023 - Apr 2023

- Frontend work, API integration and code reviews in an Agile team, with tickets tracked in Jira.

## TECHNICAL SKILLS

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**Frontend:** React, Next.js, TypeScript, Preact, Tailwind CSS, HTML/CSS

**Backend & APIs:** Python, Go, FastAPI, Gin, Flask, SQLAlchemy, Alembic, REST, OpenAPI/Swagger

**AI/ML:** RAG, Multi-Agent Systems, OpenAI, Azure OpenAI & Amazon Bedrock, LangChain, Semantic Search, Vector Embeddings, Prompt Engineering

**Cloud & DevOps:** AWS (ECS Fargate, Aurora, S3, SQS, API Gateway, Amplify, KMS, Secrets Manager, CloudWatch), Terraform, Docker, GitHub Actions

**Data & Search:** PostgreSQL, Qdrant, Typesense, Azure AI Search, Supabase, Firebase

**Security & Testing:** OAuth 2.0/OIDC, Microsoft Entra ID (MSAL), JWT, RBAC, pytest, Vitest, Sentry

## EDUCATION

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### BS Computer Science | 2019 - 2023

Laguna University

- Best in Thesis Award for KASAKA, an agricultural ML application built with Next.js, Flutter and Firebase using Naïve Bayes classification.